

Facial Feature Enhancement for Immersive Real-Time Avatar-Based Sign Language Communication using Personalized CNNs

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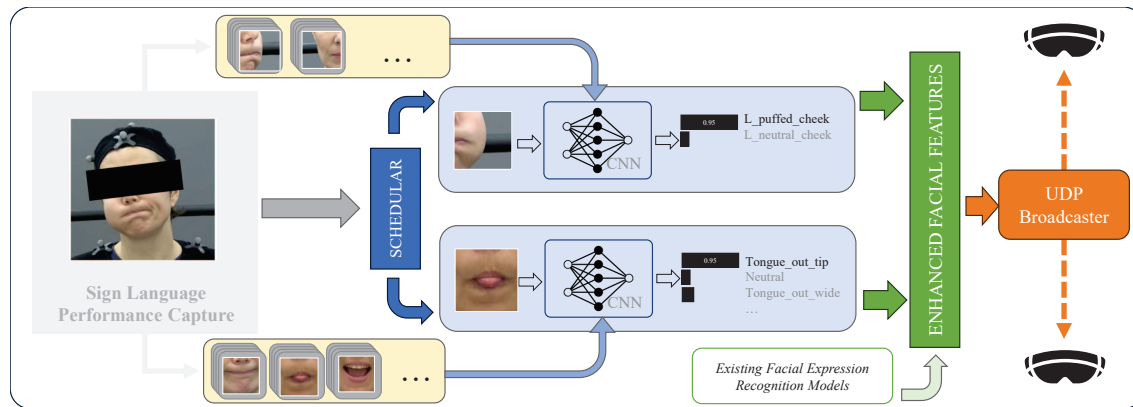


Figure 1: Our proposed approach aims to improve the recognition of special facial features through the implementation of personalized Convolutional Neural Networks (CNNs). First, the system works through an image extraction process that focuses on specific regions of the face, namely the cheeks and the mouth area in our sign language scenario. Subsequently, it performs an image classification to identify specific features in a given scheduled pattern to maintain real-time capability. Finally, our recognized movements can be integrated into existing facial recognition models and distributed over a network via UDP broadcasting.

ABSTRACT

Facial recognition is crucial in sign language communication. Especially for virtual reality and avatar-based communication, increased facial features have the potential to integrate the deaf and hard-of-hearing community to improve speech comprehension and empathy. But, current methods lack precision in capturing nuanced expressions. To address this, we present a real-time solution that utilizes personalized Convolutional Neural Networks (CNNs) to capture intricate facial details, such as tongue movement and individual puffed cheeks. Our system's classification models offer easy expansion and integration into existing facial recognition systems via UDP network broadcasting.

Index Terms: [Human-centered computing]: Accessibility—Accessibility technologies; [Computing methodologies]: Machine learning—Machine learning approaches;

1 INTRODUCTION

In the constantly evolving world of virtual reality (VR), the need for self-expression through avatars is becoming increasingly more important as they play a key role in conveying emotions and meanings, resulting in increased immersion [1, 3, 4]. Current face-tracking systems use different approaches to achieve an accurate representation of facial movements in virtual environments. Techniques such as image-based methods and machine learning can be applied to cap-

ture the most prominent facial expressions such as eyes, mouth, and lips [5]. The importance of accurate facial recognition and representation in VR is not only connected to the ability to create a holistic immersive social presence but is particularly important in the context of sign language, where facial expressions are mandatory in interpersonal communication [6]. The different levels and finer details of facial expressions are often crucial for the correct interpretation of gestures and words. For example, only the mouth expression can make a difference between various meanings, even though the same hand gesture is performed. Therefore, more facial expressiveness is essential for avatar-based sign language communication beyond the most prominent features, like cheek puffing and visible tongue positions.

In our work, we present a method to capture these particular facial expressions and transfer them to a virtual avatar. Our approach relies on trained, personalized Convolutional Neural Networks (CNNs) to perform the recognition of special areas like cheeks and mouth. Our system can be adapted to different scenarios, as individual models are only trained on specific facial regions. We achieve real-time performance with only a webcam to potentially enhance the VR experience for avatar-based sign language communication.

2 METHOD

Our base idea starts by recording the person's face in motion and separately in specific poses. Afterward, it is possible to train CNNs to classify these special movements. Our developed method is capable of hooking into existing frameworks such as *Google's Mediapipe* or *Apple Live Link Face* (see Fig. 1).

2.1 Data preparation

To identify relevant regions, we captured the person's face during specific facial expressions. Only the head was rotated to ensure

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optimal coverage while maintaining consistent facial expression. For sign language, this approach was ideal, as users mostly remain positionally stationary. For each feature, we recorded 1:30-minute videos (1080p, 60 FPS) and extracted every 3rd frame to eliminate too similar images and prevent overfitting of our subsequent machine learning approach. This process generated approximately 1500 frames for each feature. In the next step, we isolated significant regions such as cheeks and the mouth for tongue movement into sub-frames. In order to stay consistent in dimensions, those regions were detected by a combination of *openCV*'s mouth detection model and *Google's Mediapipe* face landmarks model.

In addition, we applied subtle changes to the scaling, saturation, and rotations to augment the data. For symmetrical facial expressions, such as puffed cheeks on the left and right, we further expanded the data set by flipping them horizontally.

2.2 ML Network

Our machine learning approach can be seen as a classification task, determining the visibility or absence of a specific expression. Therefore, we use the fast and well-established convolutional network architecture *AlexNet* [2] for simplicity and efficiency. Although other architectures such as Convolutional Long-Short-Term Memory Networks (ConvLSTM2D) have proven to be more suitable for sequential motion data such as ours, the manual annotation needed for all sequences led us to the practicality and efficiency of an *AlexNet*.

2.3 Performance & Discussion

Using only a webcam, our classification requires 3.2ms per prediction (Intel i7 13700k) and has a total accuracy of 93% in our scenario. Though this seems to be sufficient, it can quickly scale into unusable areas when dealing with multiple models at once, especially when maintaining VR-capable frame rates. Therefore, we split the feature classification into only one detection per frame to improve performance. In our sign language scenario, both models for the cheeks and mouth with tongue movement are processed one after the other. We used alternating threads in a round-robin principle, so each frame executes only one feature classification while keeping previous results cached. This ensures not only the real-time capability of the system, but also enables independent processing of the results, which improves overall stability and the possibility to scale easily. However, we are aware that this only works to a certain point where facial features do not change rapidly. This allows for easier implementation in various existing systems, which increases the usability of our solution. For example, we were able to extend existing facial recognition frameworks such as *Google's Mediapipe* and possibly *Apple's Live Link Face* almost seamlessly.

The usage of separate instances allows us to send the results effortlessly over the network via UDP as our approach is designed to be decoupled and can be seen as its own instance. This additionally provides streaming capabilities to send the recognized facial features directly to standalone HMDs. The integration with Meta Quest Pro's facial recognition system represents a particularly important step, as it is currently considered the easiest way to recognize facial features in virtual reality (see Fig. 2).

3 LIMITATIONS & CONCLUSION

Using personalized CNNs faces several potential challenges that need to be carefully considered. Relying only on video data for our learning approach can potentially introduce inconsistent lighting or motion blur which can significantly affect the precision of facial tracking. Consistent training data videos are therefore mandatory, especially for rare or unusual facial movements. Another important limitation of our approach is the sensitivity of the technique to the subject. The detection of certain facial movements depends largely on the particular characteristics and expressions of one individual



Figure 2: Comparison of different facial poses for puffed cheeks (top) and tongue out (bottom) on different avatars (here: *MetaHuman*, *readyplayer.me* and *Charamel GmbH*). Expression without our facial feature enhancement in small square for reference.

person. In our sign language scenario, this can be seen as an advantage, as our individualized method is very precise and easy to extend with more special features.

For future work, we plan to generalize the model to allow person-independent recognition of special facial features. We are aware that this requires a large variety of training data to successfully apply the method to different persons, but can be potentially overcome by generating highly detailed artificial facial expressions.

To conclude, we have developed a method that can easily extend existing facial recognition systems using personalized CNNs to provide more authentic avatar-based sign language communication. We see great potential for the development of future facial recognition systems that overcome these challenges to further enhance the inclusivity and effectiveness of VR in avatar-based sign language communication.

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REFERENCES

- [1] S. Kimmel, F. Jung, A. Matvienko, W. Heuten, and S. Boll. Let's face it: Influence of facial expressions on social presence in collaborative virtual reality. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, pp. 1–16.
- [2] A. Krizhevsky, I. Sutskever, and G. E. Hinton. Imagenet classification with deep convolutional neural networks. *Communications of the ACM*, 60(6):84–90, 2017.
- [3] S. Y. Oh, J. Bailenson, N. Krämer, and B. Li. Let the avatar brighten your smile: Effects of enhancing facial expressions in virtual environments. *PLoS one*, 11(9):e0161794, 2016.
- [4] D. Roth, K. Waldow, M. E. Latoschik, A. Fuhrmann, and G. Bente. Socially immersive avatar-based communication. In *2017 IEEE Virtual Reality (VR)*, pp. 259–260. IEEE, 2017.
- [5] A. V. Savin, V. A. Sablina, and M. B. Nikiforov. Comparison of facial landmark detection methods for micro-expressions analysis. In *2021 10th Mediterranean Conference on Embedded Computing (MECO)*, pp. 1–4. IEEE, 2021.
- [6] U. Von Agris, M. Knorr, and K.-F. Kraiss. The significance of facial features for automatic sign language recognition. In *2008 8th IEEE international conference on automatic face & gesture recognition*, pp. 1–6. IEEE, 2008.